

HAMI, China

WMO: 522030

Lat: 42.817N

Lon: 93.517E

Elev: 739

StdP: 92.76

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.9	-17.7	-25.7	0.4	-11.5	-23.7	0.5	-11.1	3.7	-6.4	3.4	-7.2	1.1	40	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	37.4	19.3	35.9	18.8	34.6	18.3	21.6	33.2	20.6	32.3	19.7	31.2	1.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	14.2	26.5	16.7	13.0	26.1	15.6	12.1	26.0	67.1	33.1	63.3	32.3	60.0	31.1	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.9	4.6	3.6	DB	-21.5	40.0	2.9	1.1	-23.6	40.8	-25.3	41.5	-26.9	42.1	-29.0	42.9	(4)
			WB	-21.9	23.5	2.8	1.6	-23.9	24.7	-25.6	25.6	-27.1	26.5	-29.1	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-9.5	-2.3	6.4	14.9	20.7	25.6	27.4	25.8	19.5	11.2	2.0	-7.2	(6)
(7)		DBStd	13.21	3.49	4.08	4.87	4.65	3.81	3.01	2.69	2.84	3.50	4.06	4.38	3.97	(7)
(8)		HDD10.0	1905	603	346	133	13	0	0	0	1	36	241	533		(8)
(9)		HDD18.3	3483	862	579	370	120	22	1	0	1	27	222	490	791	(9)
(10)		CDD10.0	2368	0	0	21	159	331	469	538	489	286	73	1	0	(10)
(11)		CDD18.3	904	0	0	0	17	93	219	280	232	62	1	0	0	(11)
(12)		CDH23.3	11172	0	0	2	232	1137	2677	3536	2755	782	51	0	0	(12)
(13)		CDH26.7	5630	0	0	0	52	447	1382	1989	1467	287	6	0	0	(13)
(14)	Wind	WSAvg	1.4	1.2	1.4	1.9	2.0	1.8	1.5	1.4	1.3	1.2	1.1	1.2	1.1	(14)
(15)	Precipitation	PrecAvg	38	1	1	1	3	3	7	7	5	3	3	2	1	(15)
(16)		PrecMax	79	10	10	15	20	14	33	30	22	13	19	14	15	(16)
(17)		PrecMin	9	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	17	2	2	3	4	4	9	7	5	3	4	3	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	11.5	23.2	30.7	35.0	37.7	39.7	38.7	34.5	27.5	17.1	5.1	(19)
(20)			MCWB	-2.3	2.9	9.0	13.4	16.3	18.6	20.4	19.7	17.2	12.9	7.3	0.0	(20)
(21)		2%	DB	-0.1	9.0	20.5	28.1	32.9	36.0	38.0	36.7	32.0	24.7	14.1	2.4	(21)
(22)			MCWB	-3.9	1.7	7.7	12.0	15.4	17.9	20.0	18.9	16.1	11.6	5.5	-1.9	(22)
(23)		5%	DB	-1.7	7.0	18.1	26.0	31.1	34.7	36.4	35.2	30.0	22.3	11.8	0.5	(23)
(24)			MCWB	-5.1	0.5	6.6	11.2	14.8	17.8	19.5	18.5	15.4	10.6	4.5	-3.1	(24)
(25)		10%	DB	-3.3	5.0	15.6	24.0	29.2	33.2	34.8	33.6	27.9	19.9	9.2	-1.3	(25)
(26)			MCWB	-6.2	-0.7	5.5	10.4	14.0	17.3	19.1	18.2	14.6	9.4	3.1	-4.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.1	3.3	9.5	14.1	18.2	21.6	23.4	22.7	19.0	13.5	7.5	0.3	(27)
(28)			MCDWB	1.9	10.6	21.4	28.8	31.5	32.9	34.0	33.7	29.7	25.3	16.5	4.5	(28)
(29)		2%	WB	-3.7	1.9	8.1	12.8	16.6	20.2	22.0	21.2	17.6	12.2	5.9	-1.7	(29)
(30)			MCDWB	-0.4	8.4	19.5	26.6	30.8	31.9	33.8	32.3	28.6	23.0	13.3	2.1	(30)
(31)		5%	WB	-4.9	0.7	6.9	11.7	15.5	19.2	21.1	20.2	16.5	11.0	4.6	-3.0	(31)
(32)			MCDWB	-1.9	6.6	17.6	24.7	29.2	31.1	33.2	31.2	27.5	20.8	11.3	0.3	(32)
(33)		10%	WB	-6.1	-0.6	5.6	10.7	14.4	18.4	20.2	19.3	15.4	9.9	3.3	-4.2	(33)
(34)			MCDWB	-3.4	4.6	15.3	22.9	27.6	30.4	31.9	30.5	26.4	18.9	9.0	-1.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.7	13.1	14.1	14.0	13.9	13.5	13.6	14.8	16.5	15.9	12.0	10.6	(35)
(36)			MCDBR	12.5	14.4	16.2	16.2	16.5	15.6	15.9	17.1	18.4	18.2	14.7	11.5	(36)
(37)			MCWBR	9.4	8.9	8.2	6.9	7.0	6.7	6.3	6.7	7.2	8.1	8.3	8.1	(37)
(38)		5% WB	MCDBR	12.2	13.9	15.4	14.8	15.1	13.8	14.0	15.2	16.9	17.5	14.5	11.1	(38)
(39)			MCWBR	9.2	8.6	7.9	6.2	6.8	6.6	6.4	6.8	7.2	7.9	8.2	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.334	0.382	0.428	0.474	0.477	0.488	0.480	0.449	0.426	0.400	0.364	0.334	(40)	
(41)		taud	2.105	2.016	1.912	1.861	1.905	1.958	2.018	2.083	2.066	2.046	2.084	2.110	(41)	
(42)		Ebn at Noon	765	785	796	793	803	795	796	809	793	754	718	723	(42)	
(43)		Edh at Noon	104	135	170	193	190	181	169	153	144	129	106	95	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.17	3.23	4.76	5.98	6.98	7.01	6.84	6.28	5.33	3.90	2.51	1.90	(44)	
(45)		RadStd	0.20	0.17	0.20	0.32	0.31	0.45	0.29	0.25	0.19	0.12	0.15	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.46	N/A	N/A	+0.65	-1.02	-1.46	N/A	N/A	+146	+101	(46)
(47)	Regional (3 neighbors)	+0.66	N/A	N/A	+0.76	N/A	-0.92	N/A	N/A	+146	+101	(47)

Nomenclature: See separate page